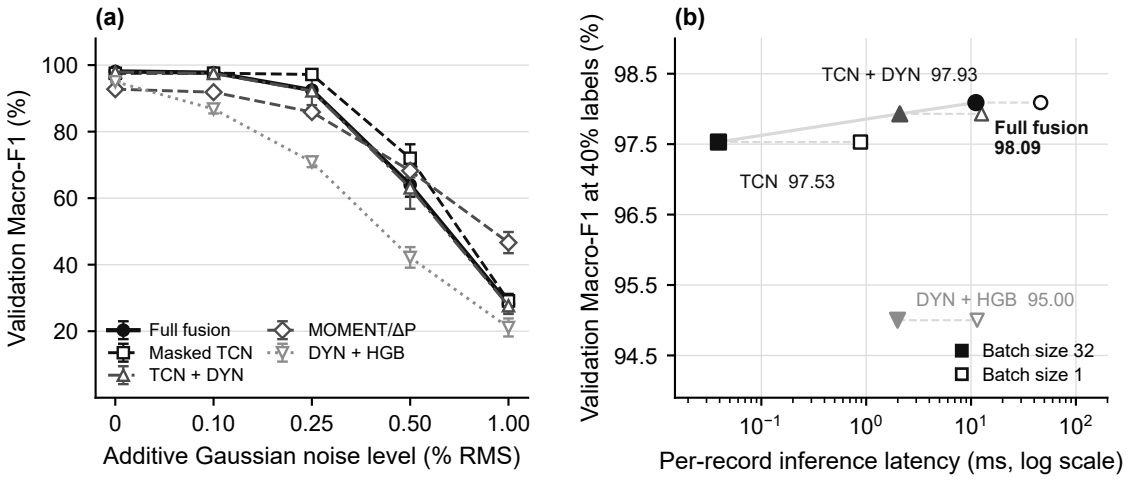

(a)
Validation Macro-F1 (%)
100
80
60
40
20
Full fusion
Masked TCN
TCN + DYN
MOMENT/ΔP
DYN + HGB
0
0.10
0.25
0.50
1.00
Additive Gaussian noise level (% RMS)
(b)
Validation Macro-F1 at 40% labels (%)
98.5
97.5
96.5
95.5
94.5
TCN + DYN 97.93
Full fusion
98.09
TCN 97.53
DYN + HGB 95.00
Batch size 32
Batch size 1
10^{-1}
10^{0}
10^{1}
10^{2}
Per-record inference latency (ms, log scale)